

Business Statistics

BBA — 3rd Semester — 2023 — Answer Sheet

Subject Code: STT201

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

Brief Answer Questions: $[10 \times 1 = 10]$

1. The covariance of X and Y is 30 and the variances of X and Y are 25 and 64 respectively. Find Karl Pearson's coefficient of correlation between X and Y.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. If the values of upper quartile and lower quartile are 63 and 59 respectively, then calculate the coefficient of quartile deviation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. The equations of two regression lines are $4X - 5Y + 33 = 0$ and $20X - 9Y - 107 = 0$, find the mean values of X and Y.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Given that $P(A \cap B) = 0.7$, $P(A) = 0.4$ and $P(B) = 0.5$, then find out the value of $P(A \cup B)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Find $P(X=4)$, the mean of Poisson distribution (λ) is 1.5.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Calculate the standard error of mean when population size (N) = 1000, sample size (n) = 100 and standard deviation (σ) = 10.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. In a moderately asymmetric distribution, the values of mean and mode are 20 and 23 respectively. Compute the value of median.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Calculate coefficient of variation, when values of mean and standard deviation are 32 and 17 respectively.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Given that value of correlation coefficient is 0.84, interpret the result on the basis of coefficient of determination.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. List out any four types of random sampling techniques.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Short Answer Questions: (Attempt any FIVE Questions)[5 × 3 = 15]

11. The following table shows the marks distribution of students in a subject:
Marks: 0-10, 10-20, 20-30, 30-40, 40-50, 50-60
Frequency: 10, 25, 45, 20, 18, 12
Find the lowest marks of the top 20% of the students.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. A random sample of 50 items gave mean weight of 7.5 kg and a standard deviation of 1.5 kg. Find 95% confidence limits of weight within which the population mean would lie.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

13. The following table represents the marks of 100 students:
Marks: 0-20, 20-40, 40-60, 60-80, 80-100
Students: 14, 27, 15
Find the missing frequencies when modal marks is 48.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

14. From the following income distribution of 500 workers in a locality, find the median value of the income distribution.
Income (000 Rs): Below 20, 20-40, 40-60, 60-80, 80-100, 100 & above
No. of workers: 50, 20, 40, 90, 40, 150, 60, 100, 50

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

15. From the following income distribution, calculate the appropriate measures of dispersion.
Daily income (Rs): Below 50, 50-100, 100-150, 150-200, 200-250, 250 and above
Number of workers: 10, 51, 100, 18, 101, 150, 25, 151, 200, 20, 16, 250 and above 11

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

16. The following information was obtained from two investment companies A and B:

	Company A	Company B
Average return (000Rs)	28	37
Variance	25	36
No. of observations	100	100

Calculate combined standard deviation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section C

Long Answer Questions: (Attempt any THREE Questions) $[3 \times 5 = 15]$

17. The following table shows the average life of electric bulbs manufactured by X and Y companies:

	Company X	Company Y
No. bulbs	n100	100
Mean life in hours	1300	1250
Standard deviation	80	90

Test whether there is any significant difference in mean life of the electric bulbs produced by two companies at 5% level significance.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

18. Daily income of the part-time staff of a bank of 50 employees was found to be normally distributed with mean of Rs 1200 and standard deviation of Rs 120. Find the probability of employees having income (a) between Rs 1050 to Rs 1400 (b) between Rs 1250 to Rs 1500.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

19. From the following distribution, find the coefficient of skewness. Also comment the result.

Monthly income (Rs 000)	Number of workers
0-100	15
100-200	50
200-300	75
300-400	40
400-500	30
500- 600	10

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

20. From the following marks distribution of students of a campus, calculate the coefficient of kurtosis. Also, interpret the result.

Marks	Below 10	10 - 20	20 - 30	30 - 40	40 - 50	50 & above
No. of students	50	100	150	90	60	50

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section D

Comprehensive Answer Questions: [20]

21. Following table shows the income and expenditure of people of certain locality of small town city of Nepal.

Income (000Rs)	Expenditure (000 Rs)
60	22
25	52
50	41
55	21
17	48
25	22
15	31
64	71
74	91
18	50
19	50

Find out:

- Coefficient of variations of income and expenditure and interpret the results.
- Correlation coefficient.
- Two regression lines.
- Estimate the expenditure of a person whose income is Rs 70 thousand.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.